SM4 DFA comparison: results and interpretation

100 paired software trials; seed 20261110. Full key verified by equality, encrypt and decrypt.

Model	Success	Mean faulty outputs	First candidates (median)	Offline median (ms)
Piret-default	100/100	32.00	2.0	0.996
Piret-matched-bits	100/100	32.00	2.0	0.975
Giraud-known-bit	100/100	32.00	2.0	1.228
Giraud-unknown-bit	100/100	35.06	10.0	4.858

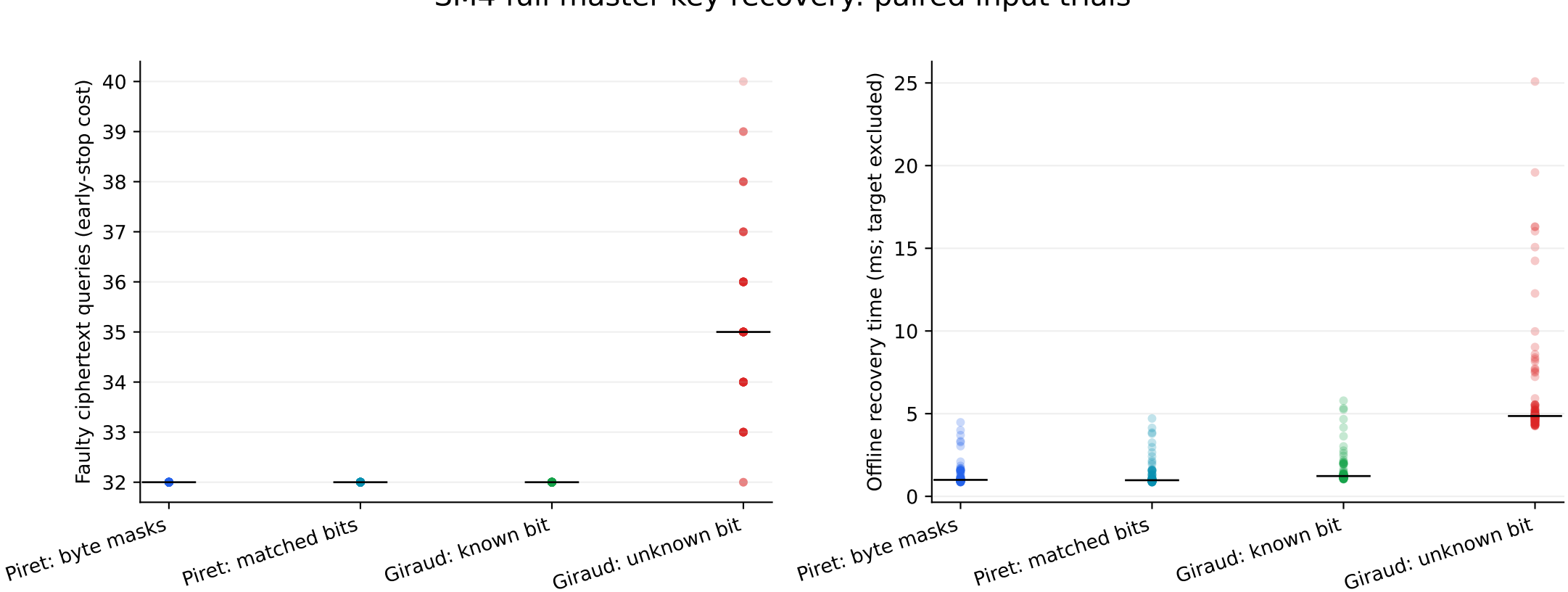
Matched Piret and known-bit Giraud produce identical complete candidate histories.

Unknown-bit Giraud uses less fault information and therefore retains more compatible inputs.

Piret-default uses byte masks; the matched-bit control isolates mask choice from solver knowledge.

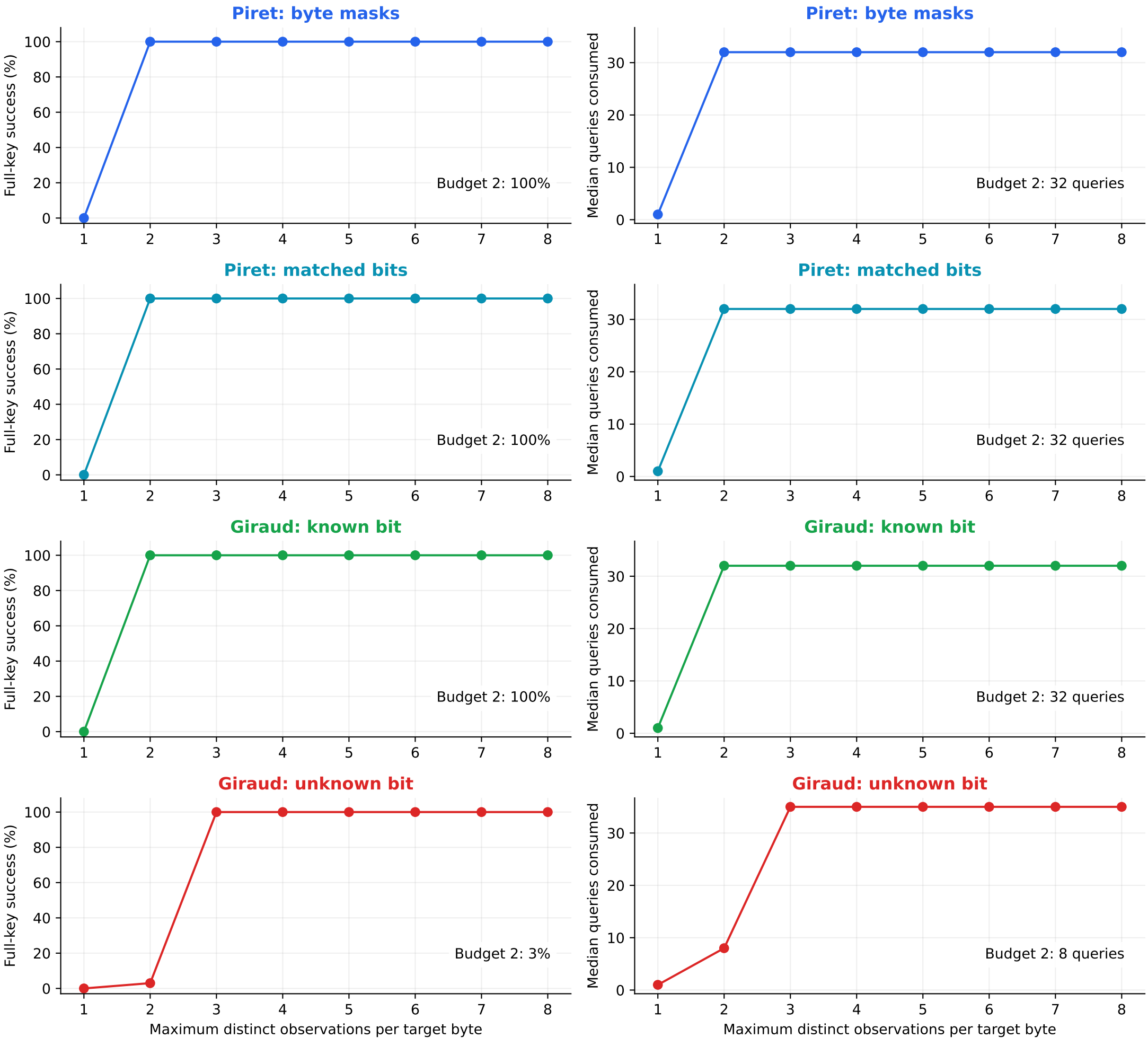
Timing excludes target encryption, serial/USB traffic and physical glitch acquisition.

All variants assume controlled faults in rounds 31, 30, 29 and 28. No STM32 experiment is measured.

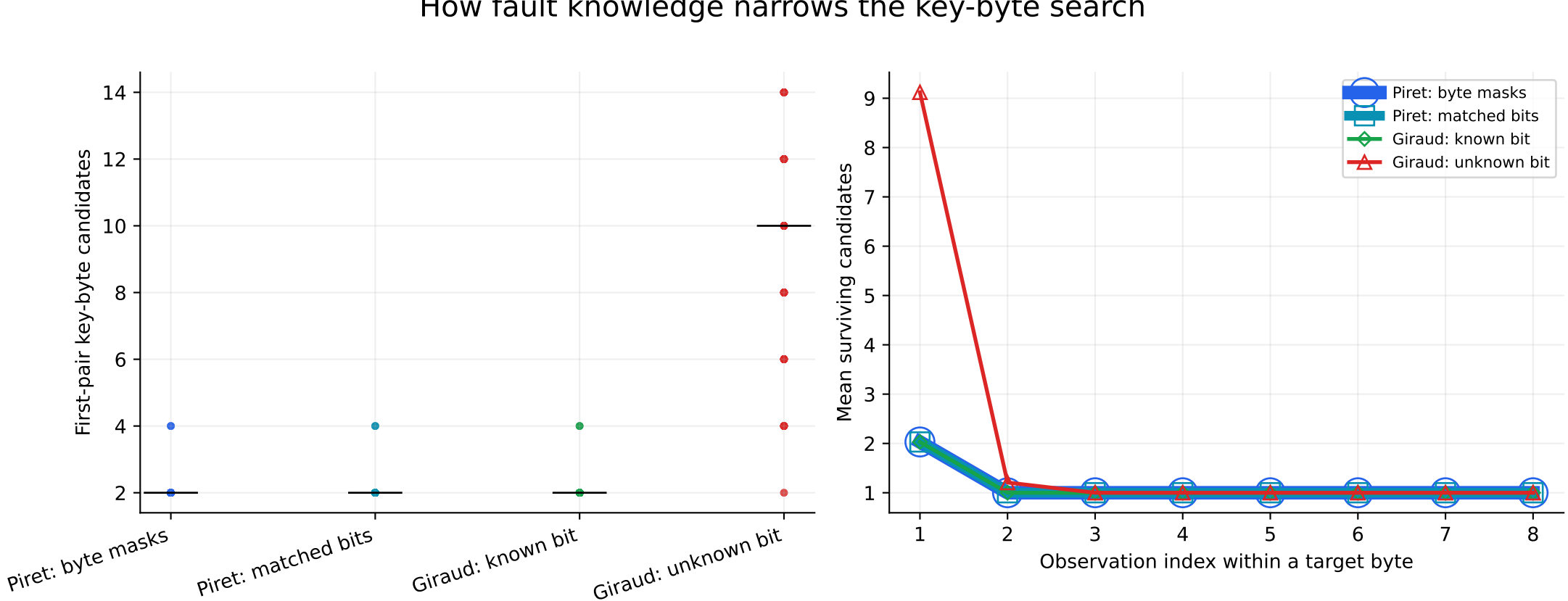


100 paired random trials. Dots: trials; black marks: medians. No hardware acquisition is timed.

Observation budget: recovery and early termination



Separate rows show every model without shifting data. Costs include incomplete attacks and early termination.


How fault knowledge narrows the key-byte search
First-pair key-byte candidates
14
12
10
8
6
4
2
Piret: byte masks
Piret: matched bits
Giraud: known bit
Giraud: unknown bit
Mean surviving candidates
9
8
7
6
5
4
3
2
1
1
2
3
4
5
6
7
8
Observation index within a target byte
Piret: byte masks
Piret: matched bits
Giraud: known bit
Giraud: unknown bit
Recovered bytes stay at one for later indices; no extra faults are queried after uniqueness.